

Homework 3

Due: September 27, 2026

1 Hidden Liquidity

Replicate Tables 2 and 3 in Avellaneda, Reedy, and Stoikov (2011), Forecasting prices from level-I quotes in the presence of hidden liquidity, *Algorithmic Finance* 1, 35-43.

- a. Pick 4 stocks out of 5.
- b. Compute the summary statistics for Table 2.
- c. Calculate Table 3: the Empirical vs. Model probabilities for the probability of an upward move on Nasdaq (T) for all 4 securities. Rows represent bid size percentiles (i), columns represent ask size percentiles (j).
- d. Hints: For table 3, look at section 4.2 of the paper describing the estimation procedure. For every bucket (i,j) count the number of future up jumps in midquote before it jumps down and divide by total number of observations in the bucket. Because the calculation is looking forward, you can use the `sign()` function on differences (`diff()`) of midquotes and count positions between up or down jumps using `which()`. For creating buckets, use `sort()` on `bidsize` and use `ceiling( length(bidsize)/10*k)` to determine positions in sorted `bidsize` for bounds between buckets. Use `nls()` for 6 to estimate H.
- e. Note the typo in the formula for the model probabilities in Table 3 in the paper! The correct formula in (14) contains the '2' in the denominator!

Please submit Table 3 for all 4 securities. Please also include your bid and ask deciles.

Write a few paragraphs describing how you might use the results.

2 RIT Simulation

Provide a write-up for the RIT ALGO Algorithmic Trading done in Discussion Session:

- a. What was your strategy going in?
- b. What happened during the simulation? What was your size strategy? Please share your Python script with an explanation on how to use it.
- c. Explain the difference of using *MARKET* Orders versus *LIMIT* Orders. What are the different risks associated with each strategy?
- d. What was your PL?

Each team member has to write its own analysis for this part.